FIVE EQUIVALENT CHARACTERIZATIONS OF GEOMETRIC REGULARITY FOR A LOCALLY NOETHERIAN SCHEME OVER A FIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and closes the cycle of implications; no step is referred to the reader.

Editorial note. The counted spans carry 3 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-geometrically-regular is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Criterion for a scheme to be regular through its local rings. Properties, Lemma properties-lemma-characterize-regular, invoked at proof lines 2446-2447. Supplies the transition from regular local rings to a regular scheme in the first implication.
- Geometric regularity of a k-algebra. Algebra, Definition algebra-definition-geometrically-regular, named in the statement and used at proof lines 2457-2459. Supplies the algebraic property to which conditions (3) and (4) are translated.
- Base change as gluing of spectra of tensor products of coordinate rings. Schemes, Section schemes-section-fibre-products, invoked at proof lines 2463-2466. Supplies the final implication from the coordinate rings of one affine covering to geometric regularity.

Source reference. Original definition 0382: sources/stacks-chow/tags.txt, source lines 4221–4221. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02IT: sources/stacks-chow/tags.txt, source lines 3312–3312. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original section 01JO: sources/stacks-chow/tags.txt, source lines 2021–2021. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Lemma 0.1. *Let k be a field. Let X be a locally Noetherian scheme over k. The following are equivalent*

(1) *X is geometrically regular,*
(2) *$X_{k'}$ is a regular scheme for every finitely generated field extension k'/k,*
(3) *$X_{k'}$ is a regular scheme for every finite purely inseparable field extension k'/k,*
(4) *for every affine open $U \subset X$ the ring $\mathcal{O}_X(U)$ is geometrically regular (see Algebra, Definition 0382), and*
(5) *there exists an affine open covering $X = \bigcup U_i$ such that each $\mathcal{O}_X(U_i)$ is geometrically regular over k.*

Proof. Assume (1). Then for every finitely generated field extension k'/k and every point $x' \in X_{k'}$ the local ring of $X_{k'}$ at x' is regular. By Properties, Lemma 02IT this means that $X_{k'}$ is regular. Hence (2).

It is clear that (2) implies (3).

Assume (3) and let $U \subset X$ be an affine open subscheme. Then $U_{k'}$ is a regular scheme for any finite purely inseparable extension k'/k (including $k = k'$). This means that $k' \otimes_k \mathcal{O}(U)$ is a regular ring for all finite purely inseparable extensions k'/k. Hence $\mathcal{O}(U)$ is a geometrically regular k-algebra and we see that (4) holds.

It is clear that (4) implies (5). Let $X = \bigcup U_i$ be an affine open covering as in (5). For any field extension k'/k the base change $X_{k'}$ is gotten by gluing the spectra of the rings $\mathcal{O}_X(U_i) \otimes_k k'$ (see Schemes, Section 01JO). Hence $X_{k'}$ is regular. So (1) holds. □